6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 1

Non-cooperative games (part I)

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Algorithmic game theory (AGT)

is the study of algorithms in strategic environments where the inputs come from self-interested agents who may manipulate the system to improve their own outcomes. It combines tools from theoretical computer science and economics to both analyze equilibrium behaviour and design mechanisms that are efficient, fair, and computationally tractable.

AGT

Resource Allocation and Scheduling

- Efficiently allocate shared resources (memory, bandwidth, tasks) among competing agents using protocols that account for strategic behavior (e.g., auctions, congestion games).
- *Example: Cloud compute allocation, energy market bidding by smart devices.*

Mechanism Design for Incentives

- Design auction or payment mechanisms that incentivize truthful behavior and maximize system efficiency.
- *Example: Online advertising bidding, spectrum auctions for wireless networks.*

AGT

Equilibrium Computation in Networked Agents

- Compute equilibria (Nash, correlated, market equilibria) to predict or guide agent behavior in competition or cooperation.
- *Example: Routing in communication networks, multiagent negotiation, security games.*

Multiagent Reinforcement Learning

- Formulate learning in interactive multiagent environments as a game, analyzing convergence and performance using equilibrium concepts.
- *Example: AI agents learning to cooperate or compete in simulated/real-world games.*

Computational Social Choice (COMSOC)

is an interdisciplinary field that studies collective decision-making and preference aggregation using tools from computer science, economics, and social choice theory. It asks how a group's individual preferences over alternatives (candidates, allocations, outcomes) can be combined into a collective decision, while explicitly accounting for computational issues. It examines both how to design such aggregation rules and how hard it is to compute their outcomes in realistic settings.

COMSOC

Collective Decision-Making

- Aggregate preferences and votes of multiple agents (human or AI) for joint decision-making (e.g., selecting actions, committee members, leaders).
- *Example: Group planning for robot teams, committee selection in organizations.*

Fair Division

- Design and implement algorithms for fair division of resources among multiple agents, guaranteeing properties like envy-freeness and proportionality.
- *Example: Allocation of workspace in office automation, fair inheritance division.*

COMSOC

Ranking and Recommendation

- Use rank aggregation/voting rules to combine agent (or user) preferences into a group-level ranking or recommendation.
- *Example: AI-based recommender systems, group-based collaborative filtering.*

Participatory Budgeting and Resource Assignment

- Agents or humans select among projects/tasks/resources; COMSOC principles ensure fairness, proportionality, and representativeness.
- *Example: Autonomous vehicles choosing charging stations, distributed sensor deployment.*

Both AGT and COMSOC together

Automated Negotiation Bargaining

- Mechanism design blends with social choice for consensus among competitive/cooperative agents.

AI Ethics Governance

- Aggregating stakeholder preferences for value alignment and ethical decision-making.

Distributed Consensus Coordination

- Joint protocols for teams of autonomous agents using game-theoretic reasoning and voting.

Multi-agent systems (MAS)

- A multi-agent system is composed of multiple interacting software components (agents), which may be *cooperative*.

MAS

- A multi-agent system is composed of multiple interacting software components (agents), which may be *competitive*.

Strategic games

- This is modelled as a **non-cooperative game**, where players *strategically optimise their own utility functions*.

Basic game-theoretic concepts

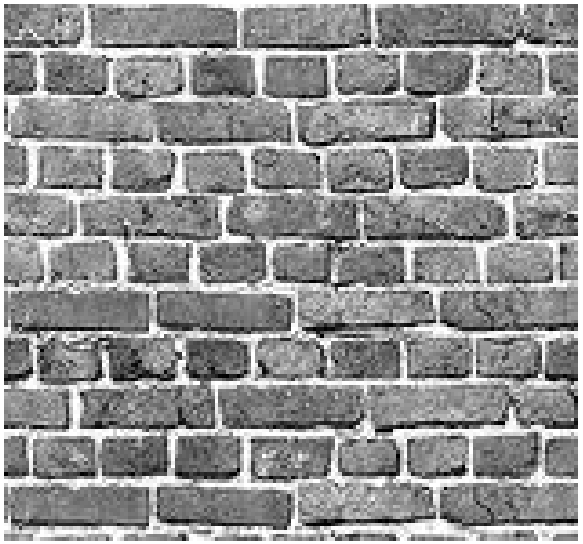
- WHAT ARE GAMES?
  - What do they model?
  - How do we represent them?
- WHAT IS A SOLUTION FOR A GAME?
  - How to play?
  - What is a reasonable outcome?

Basic game-theoretic concepts

- CAN WE COMPUTE A SOLUTION?
  - Does it always exist?
  - How many are they?
  - How (hard is it) to find them?
  - Can we expect the participants to naturally converge to a solution?
- HOW TO EVALUATE A SOLUTION?
  - How good is it comparing to other possible outcomes?

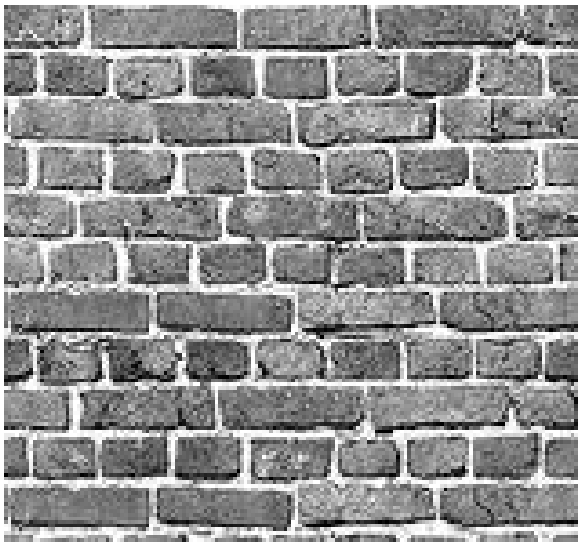
Prisoner's dilemma

- 2 suspects in a crime are interrogated in separate rooms.



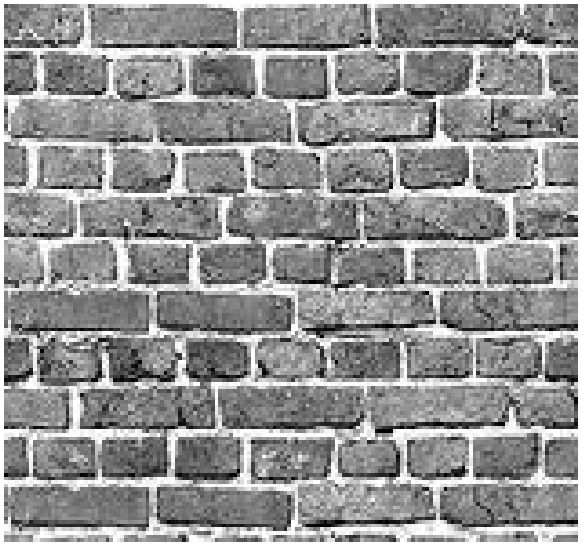
Prisoner's dilemma

- Each has 2 choices: Cooperate (deny the crime) or Defect (rat the other out).



Prisoner's dilemma

- With *no* confessions, there is enough evidence to convict on lesser charge; *one* confession is enough to establish guilt.
- Police offers plea bargain for confessing.



Prisoner's dilemma: matrix form representation

- This strategic conflict can be encoded in a *payoff matrix*:

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

- The only stable outcome is (D,D).

Normal form games

- GAMES model *strategic interactions* (or, *conflicts*) between *rational* decision makers – *players*.
- In a **non-cooperative** (or, **competitive**) game, players take actions and enjoy game outcomes individually – i.e., there are no binding contracts among the players.

Normal form games

Such a game in strategic (or, **normal**) **form** consists of:

- at least 2 PLAYERS
  - Player is a general entity: individual, company, nation, protocol, animal, etc.
- a set of STRATEGIES for each player
  - Strategies are actions which a player chooses to follow, and an *outcome* is determined by mutual choice of strategies.
- a PREFERENCE RELATION for each player over possible outcomes
  - Preference relation is modeled as a *utility* (or, *payoff*) function over the set of outcomes.

Normal form games

Formally, a game is a tuple $\Gamma = (\mathbf{N}, (S_i)_{i \in \mathbf{N}}, (U_i)_{i \in \mathbf{N}})$, where

- $\mathbf{N}$ is a (finite) set of players,
- S_i is a set of strategies for each player $i \in \mathbf{N}$, and
- $U_i : S \to \mathbb{R}$ is a utility function of $i \in \mathbf{N}$ where $S = \times_{i \in \mathbf{N}} S_i$ is the set of strategy combinations (or, *profiles*).

Normal form games

A profile s is a vector of strategies $(s_1, \ldots, s_N)$, where $s_i \in S_i$.

- We use s_{-i} to denote $(s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_N)$ – i.e., s reduced by the single entry of player i.
- Similarly, for a profile s we use s_C to denote the strategy choices of a player coalition $C \subseteq \mathbf{N}$ and s_{-C} for the complement, and we write $s = (s_C, s_{-C})$.
- Accordingly, we denote the strategy sets for player subsets.

(Some) other game types

- **extensive form** games (represented by *game trees*) where players take their actions in turns rather than simultaneously as in normal form games
- **repeated** games where some (underlying) game is played repeatedly rather than just once as in *single-shot* games, and the players strive to optimise their *overall* payoffs
- **cooperative** (or, **coalitional**) games where players, although have individual interests, coordinate their actions to achieve common goals and may have binding contracts and utility transfers (such a game is defined by a *characteristic function* over the set of possible player coalitions)

Dominance

- Notice that in PRISONER'S DILEMMA, each player is better off if they defect rather than cooperate, *no matter what the other player does*!
  - In other words, **Defect** is a DOMINANT STRATEGY for each player in this game.

Dominance

- Given a player $i \in \mathbf{N}$ and two strategies $s, t \in S_i$, we say that s **strictly dominates** t if every possible outcome when s is chosen is better for i than the corresponding outcome when t is chosen: for any $s_{-i} \in S_{-i}$, we have $U_i(s, s_{-i}) > U_i(t, s_{-i})$.
- Similarly, we say that s **weakly dominates** t if for any $s_{-i} \in S_{-i}$ we have $U_i(s, s_{-i}) \geq U_i(t, s_{-i})$ and there exists $s'_{-i} \in S_{-i}$ such that $U_i(s, s'_{-i}) > U_i(t, s'_{-i})$.

Dominance

- Strategy s is (weakly/strictly) **dominant** for i
  if it (weakly/strictly) dominates *all* $t \neq s$.
- Strategy t is (weakly/strictly) **dominanted**
  if *some* $s \neq t$ (weakly/strictly) dominates it.

Dominant strategy is a very strong solution concept
(no assumptions have to be made about other players at all!),
but does not often exists.

Dominance principle

- Rational players never choose (strictly) dominated strategies.
- Solve the game by (iteratively) eliminating dominated strategies!

Elimination of dominated strategies

- What if there are more than one dominated strategies?
  - For strong dominance, the final outcome is path-independent.
  - For weak dominance, it may depend on choices made during the elimination process.
- Surviving strategy needs not to be dominant.
- One may end up with more than one strategy per player.
  - A player may have no dominated strategies at all!

Coordination game

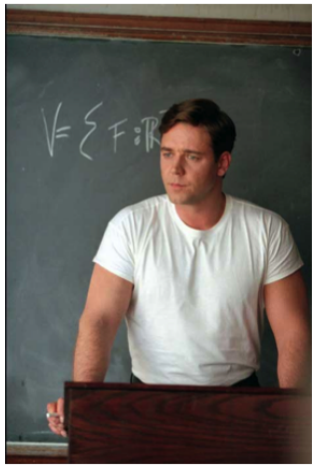
- Consider an interactive scenario where participants can realise mutual gains, but only by making consistent decisions.

PAYOFFS	**Left**	**Right**
Up	$(1, 1)$	$(0, 0)$
Down	$(0, 0)$	$(1, 1)$

- Both players benefit from choosing corresponding strategies: (**Up**, **Left**) and (**Down**, **Right**).
- No player has a dominant or dominated strategy.

Nash equilibrium

- John Nash



Nash equilibrium

- John Nash

Equilibrium

- John Nash

- Awarded 1994 Nobel Prize for Economics.

Nash equilibrium (NE)

- The concept of NE postulates the principle of *stability*.
- Intuitively, an equilibrium is a strategy profile where no one want to deviate from their strategy given other players' choices.
- Formally:
  - For a profile s, a player $i \in \mathbf{N}$ is said to have a **unilateral improving move** if there is a strategy $s_i' \in S_i$ such that $U_i(s_i', s_{-i}) > U_i(s)$.
  - A **(pure strategy) Nash equilibrium** ((PS)NE) is a profile that has no unilateral improving moves.
  - Equivalently, in an equilibrium profile s each player $i \in \mathbf{N}$ play their *best response* to the others' strategy – i.e., $U_i(s) = U_i(s_i, s_{-i}) \geq U_i(s_i', s_{-i})$ for any $s_i' \in S_i$.

Prisoner's dilemma

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

- The NE outcome is (D,D).

Coordination game

PAYOFFS	**Left**	**Right**
Up	$(1, 1)$	$(0, 0)$
Down	$(0, 0)$	$(1, 1)$

Pure strategy Nash equilibrium

- A game may have multiple PSNE.
- Or none at all!

Matching Pennies

- Consider a Matching Pennies game, where players have conflicting interests

PAYOFFS	**Head**	**Tail**
Head	$(1, -1)$	$(-1, 1)$
Tail	$(-1, 1)$	$(1, -1)$

- No pure strategy Nash equilibrium.

Mixed strategies

- A mixed strategy is a probability distribution over the set of pure strateiges.
  - A pure strategy can be viewed as mixed where only one action is played with positive probability.
  - In a general mixed strategy more than one action is played with positive probability.
- There are infinitely many mixed strategies are available to each player, even if their strategy sets are finite.

Expected payoff

- For each $i \in \mathbf{N}$, let $\Delta_i = \Delta(S_i)$ denote the set of probability distributions over S_i, and let $\Delta = \times_{i \in \mathbf{N}} \Delta_i$ be the set of all strategy profiles.
- When a mixed strategy $\tilde{s} \in \Delta$ is played, the payoff to each player is determined by their expected utility:

$$U_i(\tilde{s}) = \sum_{s \in S} U_i(s) Pr\left(s|\tilde{s}\right)$$

where $Pr\left(s|\tilde{s}\right) = \prod_{j \in \mathbf{N}} \tilde{s}_j(s_j)$ is the conditional probability of a pure action s under distribution $\tilde{s}$.

Mixed strategy Nash equilibrium

- The concept of equilibrium applies to mixed strategies in the same way as before: in an equilibrium, each player's (mixed) strategy is a best response to the strategy of the others.
- For a formal definition, replace pure actions with mixed strategies, and use expected utilities.

Mixed strategy Nash equilibrium: existence

- Mixed strategy equilibria always exist in finite games!
- For example, in MATCHING PENNIES, where there is no PSNE, $\left((\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2})\right)$ is a mixed strategy NE.

Theorem (Nash, 1951)

A game with a finite set of players and finite action sets has at least one (mixed strategy) Nash equilibrium.

Computing Mixed strategy NE in simple games

- Consider the Matching Pennies game.
  - Player 1 plays Heads with probability p.
  - Player 2 plays Heads with probability q.
- Player 2 must be *indifferent* about playing Heads or Tails – otherwise he would have chosen a pure strategy.

$$\begin{aligned} U_2(Heads) &= U_2(Tails) \\ -p + (1-p) &= p - (1-p) \\ p &= \frac{1}{2} \end{aligned}$$

Computing Mixed strategy NE in simple games

- Likewise, Player 1 must be indifferent about playing Heads or Tails.
- Thus, $p = \frac{1}{2}$ and $q = \frac{1}{2}$, and the expected utility of each player is 0.

Stag Hunt

- Stag Hunt is another generalisation of a coordination game.

PAYOFFS	**Stag**	**Hare**
Stag	$(4, 4)$	$(1, 3)$
Hare	$(3, 1)$	$(3, 3)$

- (**Stag**, **Stag**) and (**Hare**, **Hare**) are both PSNE.
- But there are also strictly mixed NE.

Stag Hunt

- Player 1 plays Stag with probability p.
- Player 2 plays Stag with probability q.

$$\begin{aligned} U_2(Stag) &= U_2(Hare) \\ 4p + (1-p) &= 3 \\ p &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} U_1(Stag) &= U_1(Hare) \\ 4q + (1-q) &= 3 \\ q &= \frac{2}{3} \end{aligned}$$

Stag Hunt

- The utility function of player 2 has changed.

PAYOFFS	**Stag**	**Hare**
Stag	$(4, 4)$	$(1, 3)$
Hare	$(3, 2)$	$(3, 3)$

Stag Hunt

- Player 1 plays Stag with probability p.
- Player 2 plays Stag with probability q.

$$\begin{aligned} U_2(Stag) &= U_2(Hare) \\ 4p + 2(1-p) &= 3 \\ p &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} U_1(Stag) &= U_1(Hare) \\ 4q + (1-q) &= 3 \\ q &= \frac{2}{3} \end{aligned}$$

Maximin (safety level) strategies

- The **maximin** (or, **safety level**) **strategy** of player i in a normal-form game is a strategy that maximises i's worst-case payoff, in the situation where all the other players happen to play the strategies which cause the greatest harm to i:

$$LS_i \in \arg\max_{s_i} \min_{s_{-i}} u_i(s_i, s_{-i})$$

  - not necessarily unique
  - in general mixed

- The **maximin value** (or, **safety level**) of the game for i is that minimum amount of payoff guaranteed by a maximin strategy:

$$u_i^{LS} = \max_{s_i} \min_{s_{-i}} u_i(s_i, s_{-i})$$

Maximin strategies: intuition

- Assume player i must commit to a (possibly mixed) strategy, and then the remaining agents $-i$ observe this strategy and choose their own strategies to minimise i's expected payoff.
- The maximin strategy is i's best choice in such situation.

Prisoner's dilemma

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

- **Defect** is a safety level strategy for each of the players.

Minimax strategies: 2-player games

- The **minimax strategy** and **minimax value** play a dual role to their maximin counterparts.
- In *two-player* games, the minimax strategy for player i against player $-i$ is a strategy that keeps the maximum payoff of $-i$ at a minimum:

$$\arg\min_{s_i} \max_{s_{-i}} u_{-i}(s_i, s_{-i})$$

  and the minimax value of player $-i$ is that minimum.
- This is useful when we want to consider the amount that one player can punish another without regard for its own payoff.
  - Such punishment can arise in repeated games.

Minimax strategies: n-player games

- In n-player games with $n > 2$, defining player i's minimax strategy against player j is a bit more complicated.
- This is because i will not usually be able to guarantee that j achieves minimal payoff by acting unilaterally.
- However, if we assume that all the players other than j choose to "gang up" on j, then we can define minimax strategies for the n-player case.
  - The players $-j$ should also be able to coordinate appropriately when there is more than one strategy profile that would yield the same minimal payoff for j.

Minimax strategies: n-player games

- In an n-player game, the **minimax strategy** for player i against player $j \neq i$ is i's component of the mixed-strategy profile s_{-j} in the expression

$$\arg\min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$$

  where $-j$ denotes the set of players other than j.
- As before, the **minimax value** for player j is

$$\min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$$

Minimax strategies: intuition

- Assume that the agents $-j$ must commit to a (possibly mixed) strategy profile, to which j can then play a best response.
- Player j receives its minimax value if players $-j$ choose their strategies in order to minimise j's expected utility after it plays its best response.

Maximin vs. minimax

- In two-player games, a player's minimax value is always equal to its maximin value.
- For games with more than two players a weaker condition holds: a player's maximin value is always less than or equal to its minimax value.

Maximin vs. minimax

- Since neither an agent's maximin strategy nor its minimax strategy depend on the strategies that the other agents actually choose, the maximin and minimax strategies give rise to solution concepts in a straightforward way.
- We will call a mixed-strategy profile $s = (s_1, s_2, \ldots, s_n)$ a *maximin strategy profile* of a given game if s_1 is a maximin strategy for player 1, s_2 is a maximin strategy for player 2 and so on.
- In two-player games, we can also define minimax strategy profiles analogously.

Maximin vs. minimax: zero-sum games

- A two-player normal-form game is **constant sum** if there exists a constant c such that for any pure strategy profile $s \in S_1 \times S_2$ it is the case that

$$u_1(s) + u_2(s) = c$$

  - For $c = 0$ it is a **zero-sum** game.
- In two-player, zero-sum games, there is a very tight connection between minimax and maximin strategy profiles.
- Furthermore, these solution concepts are also linked to the Nash equilibrium.

Maximin vs. minimax: zero-sum games

Theorem (Minimax theorem (von Neumann, 1928))

For any finite, two-player, zero-sum game the maximum value of the minimum expected gain for one player is equal to the minimum value of the maximum expected loss for the other; moreover each player has a mixed strategy which realises this equality.

Minimax theorem

- Each player's maximin value is equal to its minimax value. By convention, the maximin value for player 1 is called the **value of the game**.
- For both players, the set of maximin strategies coincides with the set of minimax strategies; and
- Any maximin strategy profile (or, equivalently, minimax strategy profile) is a Nash equilibrium. Furthermore, these are all Nash equilibria. Consequently, all Nash equilibria have the same payoff vector (namely, the one in which player 1 gets the value of the game).

Minimax theorem: Matching Pennies

PAYOFFS	**Head**	**Tail**
Head	$(1, -1)$	$(-1, 1)$
Tail	$(-1, 1)$	$(1, -1)$

- The value of the game is 0.
- The unique Nash equilibrium consists of both players randomising between heads and tails with equal probability, which is both the maximin strategy and the minimax strategy for each player.

Computing maximin and minimax strategies in a general-sum 2-player game

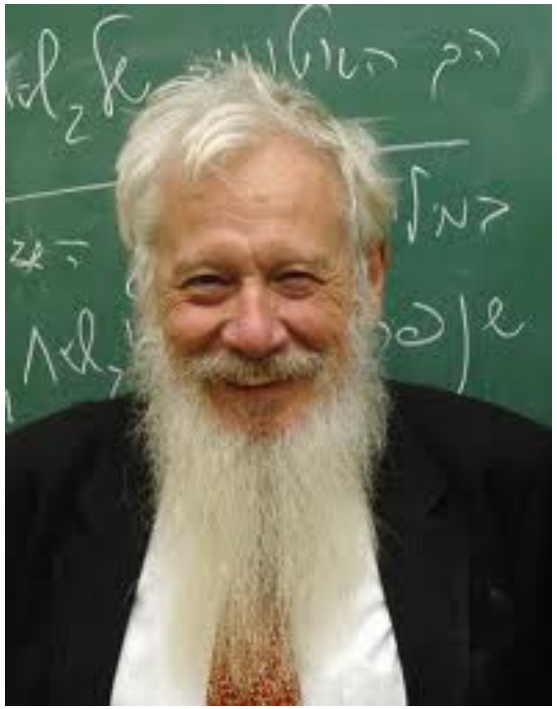
- Notice that i's maximin strategy depends only on i's utilities
  - We can change $-i$ utilities without changing i's maximin strategy.
- Replace the game with a zero-sum game where i has utilities as in the original game, and $-i$— their negatives.
- By the Minimax theorem, the NE strategies are also maximin strategies.
- Similarly, solve for minimax strategies.

Further solution concepts for normal-form games

- Rationalisable strategies
- Correlated equilibrium
- ϵ-Nash equilibrium
- Strong equilibrium

Strong equilibrium

- NE is vulnerable to deviations by coalitions of players.
- This issue is addressed in the solution concept known as *strong equilibrium*.
- Was introduced by Robert Aumann.
  - Awarded 2005 Nobel Prize for Economics.



Strong equilibrium

- For a profile s, a coalition of players $C \subseteq \mathbf{N}$ is said to have a **improving move** if there is a joint strategy $s'_C \in S_C$ such that for each $i \in C$, $U_i(s'_C, s_{-C}) > U_i(s)$.
- A **strong equilibrium** (SE) is a profile that has no improving moves for any coalition $C \subseteq \mathbf{N}$ of players.
- Obviously, every strong equilibrium is also a Nash equilibrium.

Example: Prisoner's dilemma

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

- The unique PSNE is not strong equilibrium.
- No SE at all!

Coordination game

PAYOFFS	**Left**	**Right**
Up	$(1, 1)$	$(0, 0)$
Down	$(0, 0)$	$(1, 1)$

Both PSNE are strong equilibria!

THANK YOU!